

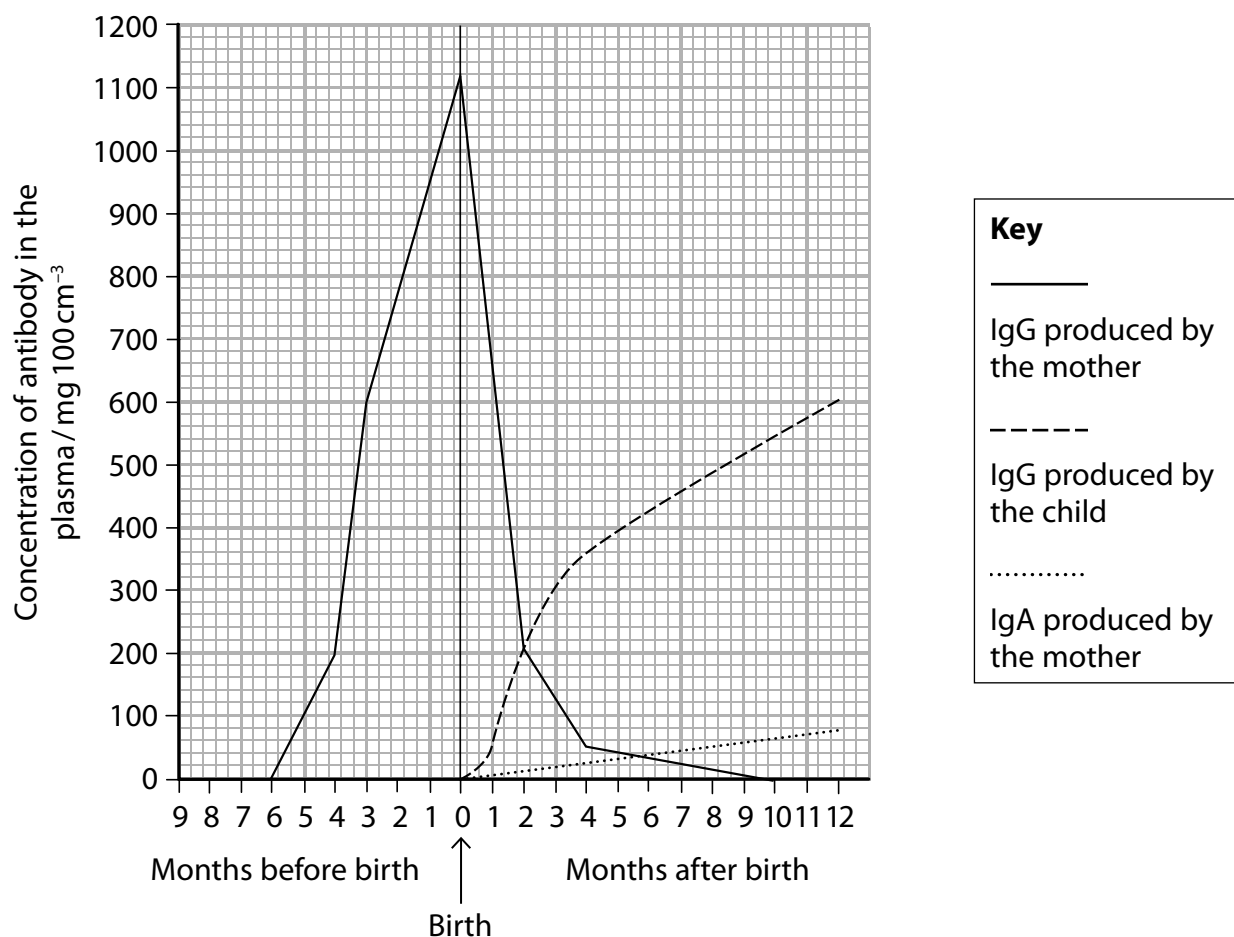
2 Antibodies play an important role in the immune response.

(a) Explain the importance of antibodies.

(4)

(b) There are different classes of antibody, including IgA, IgG and IgM.

The graph shows the changes in the concentration of IgA and IgG in the plasma of a fetus before birth and in a child after birth.



- (i) Describe the changes in the classes of antibody in a fetus before birth and in a child after birth. Use the information in the graph to support your answer.

(3)

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- (ii) Which type of immunity explains the presence of IgA?

(1)

- ☐ **A** artificial active
- ☐ **B** artificial passive
- ☐ **C** natural active
- ☐ **D** natural passive

(iii) Suggest why the concentration of antibodies is less in a child 12 months after birth than in an adult.

(2)

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(Total for Question 2 = 10 marks)